

Vansh Singh

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EDUCATION

Isenberg School of Management at UMass Amherst

Master of Science in Business Analytics

Amherst, MA

Graduating May 2027

University of Massachusetts Amherst, Riccio College of Engineering

Bachelor of Science in Computer Engineering

Amherst, MA

Graduated May 2026

PROFESSIONAL EXPERIENCE

University of Massachusetts Amherst IT

Cloud Engineering Intern

Amherst, MA

September 2025 – Present

- Eliminated 3+ hours of manual server configuration per release by building Bitbucket Pipelines CI/CD workflows and Python provisioning scripts for internal infrastructure.
- Standardized deployments across IT teams with modular Docker application templates, and contributed to Terraform and Kubernetes configurations.
- Owned Jira tickets from backlog through release within Agile sprints, tracking changes in Bitbucket and documenting deployment procedures in Confluence.
- Delivered technical initiatives from requirements discovery to production rollout, authoring architecture documentation and presenting KPI metrics to IT stakeholders.

Stem Studio

Engineering Instructor

Chicago, IL

May 2023 – Present

- Remediated a production WordPress security breach on a client-facing site, eliminating malicious remote loggers and URL redirection attacks via SQL cleanup and server diagnostics.
- Instructed 50+ students in applied computing and hardware, designing curricula covering Python, IoT sensors, microcontrollers, and circuit diagnostics.

CodeDay

Technical Mentor & Hackathon Organizer

Chicago, IL

August 2020 – August 2023

- Co-organized citywide hackathons for 150+ student developers; served on technical judging panels evaluating 30+ team software prototypes across architecture, code quality, and execution.
- Mentored 20+ student engineering teams across underserved communities to build functional Python, JavaScript, and AI applications.

LEADERSHIP AND RELEVANT PROJECTS

ASME UMass Mechatronics Team | *SolidWorks CAD, CNC Machining, Rapid Prototyping*

September 2024 – Present

- Directed a multidisciplinary subteam of 3 electrical and computer engineers to CAD-model and CNC-fabricate an articulated rover and remote payload delivery drone.
- Awarded 2nd place globally at the ASME IAM3D competition representing UMass Amherst against 40+ universities, optimizing structural load-to-weight ratios and component durability.

Quantum Hybrid Neural Network | *PyTorch, PennyLane, AWS Braket* | [GitHub](#)

November 2025 – Present

- Engineered a scalable hybrid quantum-classical neural network framework combining convolutional layers with a Quantum Mixture of Experts (MoE) via PennyLane and AWS Braket.
- Reduced model training latency on local hardware by developing an offline quantum feature caching pipeline that eliminated redundant circuit simulations during backpropagation.

U.S. Department of Energy – Collegiate Wind Competition | *Arduino, Power Electronics*

September 2025 – Present

- Directed a 2-engineer electrical subteam in designing a 6-state Arduino finite state machine and rectified DC power architecture for an offshore wind turbine.
- Engineered active voltage regulation circuits and emergency braking relays, achieving peak electrical output of 128W during wind tunnel load validation.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Java, JavaScript, Bash, SystemVerilog

Cloud & DevOps: AWS (EC2, S3, Lambda, Braket), GCP, Docker, Kubernetes, Terraform, Bitbucket Pipelines (CI/CD), Git, Linux

Machine Learning & Data: PyTorch, PennyLane, Qiskit, ETL Pipelines, Data Modeling, Statistical Analysis

Embedded & Hardware: ESP32, ARM Cortex, Arduino, I2C/SPI, IoT Sensors, Power Electronics, LTSpice, SolidWorks, CNC

Collaboration Tools: Atlassian Suite (Jira, Confluence, Bitbucket), Agile/Scrum